Product Requirements Document (PRD)

Create a modern travel and transportation marketplace website called "TravelConnect".

Project Vision

Build a platform that helps travelers find transportation options in areas where government buses are limited, schedules are unavailable, or public transport is unreliable. The platform should connect local transport providers with travelers and offer the cheapest and fastest route recommendations.

Problem Statement

Many rural and semi-urban areas lack proper transportation information. Travelers struggle to find:

- Bus timings
- Local transport operators
- Affordable cab services
- Auto-rickshaw services
- Reliable route information

The platform should solve these issues by aggregating all transport options in one place.

Target Users

1. Tourists
2. Daily commuters
3. Students
4. Rural residents
5. Local transport operators
6. Travel agencies

Core Features

1. Route Planner

- Enter source and destination.
- Show:
    - Fastest route
    - Cheapest route
    - Recommended route
- Display estimated travel time.

- Display total cost.
- Show map visualization.

2. Local Bus Services

- List private and local buses.
- Show operator details.
- Show contact number.
- Show boarding points.
- Show available timings.
- Allow operators to update schedules.

3. Cab Booking

- Similar to Uber/Ola.
- Compare prices from multiple providers.
- Live driver tracking.
- Booking management.

4. Auto-Rickshaw Services

- Nearby auto availability.
- Fare estimation.
- Booking request system.
- Driver ratings and reviews.

5. Transport Operator Portal

- Register as bus operator, cab operator, or auto driver.
- Manage routes.
- Update prices.
- Upload schedules.
- View bookings and earnings.

6. Smart Route Engine

- Analyze:
    - Cost
    - Distance
    - Travel time
    - Traffic conditions
- Recommend:
    - Cheapest route
    - Fastest route
    - Best value route

7. User Accounts

- Sign up/Login
- Google Authentication
- Booking history
- Saved routes
- Favorite operators

8. Reviews & Ratings

- Rate drivers and operators.
- Leave feedback.
- Report issues.

9. Emergency & Safety

- SOS button.
- Emergency contacts.
- Trip sharing.
- Driver verification.

10. Payment Gateway

- UPI
- Credit/Debit Cards
- Net Banking
- Wallet Payments

Admin Dashboard

- User management
- Operator approval
- Route management
- Analytics dashboard
- Revenue tracking
- Complaint handling

Tech Stack

Frontend:

- Next.js
- React
- TypeScript
- Tailwind CSS
- Shadcn UI

Backend:

- Node.js
- Express.js
- PostgreSQL
- Prisma ORM

Maps & Navigation:

- Google Maps API
- OpenStreetMap
- Mapbox

Authentication:

- Clerk or Firebase Auth

Payments:

- Razorpay
- Stripe

Cloud:

- AWS or Vercel

AI Features

- AI route recommendation.
- Demand prediction.
- Fare prediction.
- Travel assistant chatbot.
- Personalized transport suggestions.

Monetization

- Booking commission.
- Premium operator listings.
- Advertisements.
- Subscription plans for operators.

Success Metrics

- Number of active users.
- Number of registered operators.
- Bookings per month.
- Route searches per day.
- Revenue generated.
- User satisfaction rating.

UI/UX Requirements

- Mobile-first design.
- Fast loading.
- Dark/Light mode.
- Multi-language support.
- Responsive on all devices.
- Clean modern travel-themed interface.

Generate:

1. Complete system architecture.
2. Database schema.
3. User flow diagrams.
4. Wireframes.
5. API documentation.
6. Admin dashboard design.
7. MVP roadmap.
8. Development timeline.
9. Cost estimation.
10. Full implementation plan.

Technical Requirements Document (TRD)

Project Name

TravelConnect – Smart Travel & Transportation Marketplace

Overview

TravelConnect is a web-based platform that aggregates local transportation services such as buses, cabs, auto-rickshaws, and shared vehicles, especially in regions with limited public transport availability. The platform provides route optimization, fare comparison, booking management, and operator onboarding.

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1. System Architecture

Architecture Type:

- Microservices-ready Modular Monolith (MVP)
- REST API Architecture
- Cloud-Native Deployment

Components:

1. Frontend Application
2. Backend API Server
3. Authentication Service
4. Route Optimization Engine
5. Booking Service
6. Payment Service
7. Notification Service
8. Admin Portal
9. Database
10. Maps & Geolocation Service

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2. Technology Stack

Frontend:

- Next.js 15
- React 19
- TypeScript
- Tailwind CSS
- Shadcn/UI
- Redux Toolkit

Backend:

- Node.js
- Express.js
- TypeScript

Database:

- PostgreSQL

ORM:

- Prisma

Authentication:

- Firebase Auth or Clerk

Maps:

- Google Maps API
- OpenStreetMap Backup

Payment:

- Razorpay
- Stripe

Notifications:

- Firebase Cloud Messaging
- Twilio SMS
- Email Service

Hosting:

- Vercel (Frontend)
- AWS EC2/ECS (Backend)
- AWS RDS PostgreSQL

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3. User Roles

Customer

- Register/Login
- Search routes
- Book transportation
- Make payments
- Track bookings
- Review operators

Operator

- Register business
- Manage routes
- Manage schedules
- View bookings
- Manage pricing

Driver

- Accept rides
- Update trip status
- Navigation support

Admin

- Approve operators
- Manage users

- Handle disputes
- View analytics

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4. Functional Requirements

Authentication Module

Features:

- Email registration
- Mobile OTP login
- Google Sign-In
- Password reset
- JWT Authentication

Security:

- MFA support
- Refresh tokens
- Session management

Route Planner Module

Input:

- Source
- Destination
- Travel Date

Output:

- Cheapest Route
- Fastest Route
- Recommended Route

Route Factors:

- Distance
- Fare
- Travel Time
- Traffic Data
- Vehicle Availability

Bus Service Module

Features:

- Bus Listings
- Schedule Management
- Seat Availability
- Fare Information
- Boarding Locations

Cab Service Module

Features:

- Ride Booking
- Fare Calculation
- Driver Assignment
- Live Tracking

Auto Service Module

Features:

- Nearby Auto Discovery
- Fare Estimate
- Instant Booking

Booking Management

Features:

- Create Booking
- Cancel Booking
- Booking History
- Booking Status

Statuses:

- Pending
- Confirmed
- In Progress

- Completed
- Cancelled

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5. Database Design

Users

- id
- name
- email
- phone
- role
- created_at

Operators

- id
- user_id
- business_name
- verification_status

Drivers

- id
- operator_id
- vehicle_number
- license_number

Routes

- id
- source
- destination
- distance
- duration

Vehicles

- id
- operator_id
- type
- capacity

Bookings

- id

- user_id
- vehicle_id
- route_id
- amount
- status

Payments

- id
- booking_id
- transaction_id
- payment_status

Reviews

- id
- user_id
- operator_id
- rating
- comment

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6. API Requirements

Authentication:

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/logout

Routes:

- GET /api/routes/search
- GET /api/routes/recommend

Bookings:

- POST /api/bookings
- GET /api/bookings/:id
- DELETE /api/bookings/:id

Operators:

- POST /api/operators/register
- PUT /api/operators/update

Payments:

- POST /api/payments/create
- POST /api/payments/verify

Reviews:

- POST /api/reviews
- GET /api/reviews

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7. Non-Functional Requirements

Performance:

- Page Load < 2 sec
- API Response < 500 ms

Scalability:

- 100,000+ users
- 10,000 concurrent users

Availability:

- 99.9% uptime

Security:

- HTTPS
- JWT
- Data Encryption
- SQL Injection Protection
- XSS Protection

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8. AI Recommendation Engine

Inputs:

- User Location
- Destination
- Traffic Data
- Fare Data

Outputs:

- Cheapest Route
- Fastest Route
- Best Route Score

ML Models:

- Route Recommendation
- Demand Forecasting
- Fare Prediction

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9. Admin Dashboard

Features:

- User Analytics
- Booking Analytics
- Revenue Dashboard
- Operator Approval
- Complaint Management

Reports:

- Daily Revenue
- Monthly Revenue
- Active Operators
- Popular Routes

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10. Deployment Pipeline

Development:

- GitHub

CI/CD:

- GitHub Actions

Deployment:

- Vercel
- AWS

Monitoring:

- Sentry
- Prometheus
- Grafana

Logging:

- Winston
- CloudWatch

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11. Future Enhancements

Phase 2:

- Android App
- iOS App
- Multilingual Support

Phase 3:

- AI Travel Assistant
- Dynamic Pricing
- Voice Search

Phase 4:

- Intercity Travel Marketplace
- Hotel Booking Integration
- Tourism Packages

Generate detailed:

- ER Diagram
- Database Schema
- API Swagger Documentation
- System Design Diagram
- Low-Level Design (LLD)
- High-Level Design (HLD)
- Kubernetes Deployment Architecture
- Security Architecture

TravelConnect - App Flow

1. Splash Screen

- App Logo
- Tagline: "Find the Best Way to Travel"

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2. Welcome Screen

Options:

- Login
- Register
- Continue as Guest

↓

3. Select User Type

Customer

OR

Service Provider

Provider Types:

- Bus Operator
- Cab Driver
- Auto Driver
- Bike Taxi Driver
- Travel Agency
- Shuttle Service Provider

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4. Registration / Login

Customer Registration

- Name
- Mobile Number
- Email
- Password
- Google Login

Provider Registration

- Name
- Mobile Number
- Vehicle Type
- Vehicle Number
- License Number
- Business Details
- Verification Documents

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5. Home Screen

Top Section:

- Current Location
- Profile Icon
- Notifications

Search Box: "Where do you want to go?"

Fields: 📍 Source Location 🏁 Destination Location

Options:

- Current Location Button
- Map Selection

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6. Map Screen

Google Maps/OpenStreetMap Integration

Features:

- Pin Source Location
- Pin Destination Location
- Distance Calculation
- Estimated Time

Button: "Find Transport"

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7. Transport Options Screen

Show all available transport modes:

🚌 Bus 🚖 Cab 🏍️ Bike Taxi 🚂 Train ✈️ Flight 🛺 Auto Rickshaw 🚐 Shared Vehicle

For each service show:

- Starting Price
- Travel Time
- Distance
- Number of Providers Available

Filters:

- Cheapest
- Fastest
- Best Rated
- Nearest Pickup

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8. Service Providers List

Example: User clicks "Cab"

Show:

Provider Card

Provider Name Vehicle Type Price Rating Estimated Arrival Time

Examples:

- Uber Cab
- Local Cab Service
- City Taxi
- FastRide

User can compare prices.

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9. Provider Details Page

Show:

- Provider Name
- Vehicle Information
- Driver Details
- Ratings & Reviews

- Pickup Location
- Route Map
- Estimated Fare
- Estimated Time

Buttons:

- Book Now
- Contact Driver

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10. Booking Confirmation

Show:

- Source
- Destination
- Provider
- Fare
- Taxes
- Total Amount

Payment Methods:

- UPI
- Credit Card
- Debit Card
- Wallet

Button: Confirm Booking

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11. Live Tracking Screen

Show:

- Driver Location
- Route Map
- ETA
- Driver Contact

Options:

- Share Trip
- SOS Button
- Cancel Ride

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12. Trip Completed

Show:

- Trip Summary
- Distance
- Time
- Fare

Options:

- Rate Driver
- Leave Review

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CUSTOMER PROFILE

Profile Page

- Personal Details
- Booking History
- Saved Routes
- Favourite Providers
- Payment Methods
- Notifications

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SETTINGS

- Language
- Dark Mode
- Location Settings
- Privacy Settings
- Notification Settings

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HELP & SUPPORT

- FAQs
- Live Chat
- Call Support

- Report Issue
- Refund Request

SERVICE PROVIDER FLOW

Login/Register

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Dashboard

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Manage Services

- Add Vehicle
- Add Route
- Add Schedule
- Set Price

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Bookings Screen

- New Bookings
- Ongoing Trips
- Completed Trips

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Earnings Screen

- Daily Earnings
- Weekly Earnings
- Monthly Earnings

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Profile & Verification

- Upload Documents
- Vehicle Details
- Bank Account Details

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Support Center

- Contact Admin
- Raise Ticket

Bottom Navigation Bar

🏠 Home

🗺️ Routes

📖 Bookings

🔔 Notifications

👤 Profile

TravelConnect Backend Database Schema (SQL)

1. Users Table

Stores all customers and service providers.

CREATE TABLE users (

id BIGINT PRIMARY KEY AUTO_INCREMENT,

full_name VARCHAR(100) NOT NULL,

email VARCHAR(150) UNIQUE,

phone VARCHAR(20) UNIQUE NOT NULL,

password_hash VARCHAR(255) NOT NULL,

role ENUM(

'customer',

'cab_driver',

'bus_operator',

'auto_driver',

```
        'bike_driver',
        'train_provider',
        'flight_provider',
        'admin'
    ) NOT NULL,
    profile_image VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

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2. Service Providers Table

Additional information for providers.

```
CREATE TABLE providers (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    user_id BIGINT NOT NULL,
    business_name VARCHAR(150),
    license_number VARCHAR(100),
    verification_status ENUM(
        'pending',
        'approved',
        'rejected'
    ) DEFAULT 'pending',
    rating DECIMAL(3,2) DEFAULT 0,
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

3. Vehicles Table

```
CREATE TABLE vehicles (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    provider_id BIGINT NOT NULL,
    vehicle_type ENUM(
        'cab',
        'bus',
        'auto',
        'bike',
        'train',
        'flight'
    ),
    vehicle_number VARCHAR(50),
    model_name VARCHAR(100),
    capacity INT,
    status ENUM(
        'available',
        'busy',
        'inactive'
    ) DEFAULT 'available',
    FOREIGN KEY (provider_id) REFERENCES providers(id)
);
```

4. Locations Table

```
CREATE TABLE locations (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    location_name VARCHAR(255),
    latitude DECIMAL(10,8),
    longitude DECIMAL(11,8)
);
```

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5. Routes Table

```
CREATE TABLE routes (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    provider_id BIGINT,
    source_location_id BIGINT,
    destination_location_id BIGINT,
    distance_km DECIMAL(10,2),
    estimated_time_minutes INT,
    FOREIGN KEY (provider_id) REFERENCES providers(id),
    FOREIGN KEY (source_location_id) REFERENCES locations(id),
    FOREIGN KEY (destination_location_id) REFERENCES locations(id)
);
```

---

6. Pricing Table

Different providers can offer different prices.

```
CREATE TABLE pricing (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    route_id BIGINT,
    vehicle_id BIGINT,
    base_fare DECIMAL(10,2),
    price_per_km DECIMAL(10,2),
    surge_multiplier DECIMAL(5,2),
    FOREIGN KEY (route_id) REFERENCES routes(id),
    FOREIGN KEY (vehicle_id) REFERENCES vehicles(id)
);
```

---

7. Bookings Table

```
CREATE TABLE bookings (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    customer_id BIGINT,
    provider_id BIGINT,
    vehicle_id BIGINT,
    source_location_id BIGINT,
    destination_location_id BIGINT,
    total_price DECIMAL(10,2),
    booking_status ENUM(
        'pending',
        'accepted',
        'ongoing',
        'completed',
```

```
    'cancelled'
  ),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES users(id),
  FOREIGN KEY (provider_id) REFERENCES providers(id),
  FOREIGN KEY (vehicle_id) REFERENCES vehicles(id)
);
```

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8. Payments Table

```
CREATE TABLE payments (
  id BIGINT PRIMARY KEY AUTO_INCREMENT,
  booking_id BIGINT,
  amount DECIMAL(10,2),
  payment_method VARCHAR(50),
  payment_status ENUM(
    'pending',
    'success',
    'failed'
  ),
  transaction_id VARCHAR(255),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (booking_id) REFERENCES bookings(id)
);
```

---

9. Reviews Table

```
CREATE TABLE reviews (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    booking_id BIGINT,
    customer_id BIGINT,
    provider_id BIGINT,
    rating INT,
    review TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES bookings(id)
);
```

---

10. Support Tickets Table

```
CREATE TABLE support_tickets (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    user_id BIGINT,
    subject VARCHAR(255),
    message TEXT,
    status ENUM(
        'open',
        'in_progress',
        'resolved'
    ) DEFAULT 'open',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Data Flow

Registration

User → API → Users Table

Provider → API → Users Table → Providers Table

Search Route

User enters:

- Source
- Destination

API checks:

Locations Table → Routes Table → Vehicles Table → Pricing Table

Returns:

- Cheapest Route
- Fastest Route
- All Available Providers

Booking

User clicks Book

Data stored in: Bookings Table

↓

Payment completed

Data stored in: Payments Table

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Booking status updated

Review

Customer submits review

Data stored in: Reviews Table

Provider rating recalculated

Admin Access

Admin can access:

- Users
- Providers
- Vehicles
- Routes
- Pricing
- Bookings
- Payments
- Reviews
- Support Tickets

through a secure admin dashboard.